Segment Anything Model (SAM) on Cityscapes

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1 Abstract

We conducted a study to assess the capabilities of the Segment Anything Model (SAM) in the specialized domain of urban driving segmentation, which is a critical task for autonomous vehicles. We used a two pronged approach using the Cityscapes dataset. First, we established a zero-shot baseline using bounding box prompts to guide the pretrained SAM. Secondly, we evaluated the effect of minimal domain adaptation. For the zero-shot task, we observed a mean Intersection over Union (mIoU) of 0.7192 across five key classes (road, sidewalk, person, car, and building). For the adaptation, we froze SAM's ViT-H image encoder and trained a lightweight semantic segmentation head consisting of two convolutional layers on Cityscapes ground truth labels. This adaptation gave us a significant performance increase, achieving an mIoU of 0.8013, representing an 8.21 percentage point improvement. Our results show that while SAM already provides high general purpose efficiency, a minimal or lightweight task specific semantic head is highly effective and can be implemented for achieving even better performance on specific downstream tasks. However, the zero-shot SAM's poor performance is mostly due to the difficulty in providing SAM with quality prompts.

Code for the box prompt zero-shot SAM implementation and the Cityscapes-tuned head implementation is available at:

- https://github.com/derekowen-dev/SAM-for-Cityscapes

The demo video link is available here:

- https://www.youtube.com/watch?v=YBVnFD_mBps

Code for the initial point prompt zero-shot implementation is available at:

- https://github.com/erik-p04/SAM-Cityscapes-Project

2 Introduction/Background

Image segmentation is a fundamental task in computer vision that involves partitioning images into meaningful regions or objects. This capability is essential for numerous applications, including autonomous driving, medical image analysis, augmented reality, and content-based image editing. However, traditional segmentation models face significant limitations: they are typically designed for specific tasks, require extensive task-specific training with expensive pixel-level annotations, and struggle to generalize beyond their narrow training domains (Kirillov et al., 2023).

The Segment Anything Model (SAM), introduced by Meta AI, represents a paradigm shift in approaching segmentation tasks. Unlike conventional models optimized for particular datasets or object categories, SAM was designed as a foundation model for image segmentation—a single, promptable model capable of segmenting arbitrary objects in diverse images without task-specific fine-tuning. By training on the massive SA-1B dataset containing 1.1 billion masks across 11 million images, SAM learns generalizable visual representations that enable zero-shot transfer to novel segmentation tasks (Kirillov et al., 2023).

Despite SAM's impressive zero-shot capabilities, questions remain about its performance on specialized domains that differ substantially from its training distribution. Urban driving scenes, for instance, present unique challenges including complex occlusions, diverse lighting conditions, and safety-critical object classes. The Cityscapes dataset has emerged as a standard benchmark for evaluating segmentation models on such scenarios, providing high-quality pixel-level annotations of urban street scenes across 50 cities (Cordts et al., 2016).

This study investigates two complementary approaches to applying SAM to urban driving segmentation. First, we evaluate SAM's zero-shot performance on Cityscapes using box prompts, measuring how well the model generalizes to this domain without any adaptation. Second, we explore whether lightweight domain-specific training can improve upon zero-shot performance by training a semantic segmentation head on top of SAM's frozen image encoder. This two-pronged approach allows us to assess both the breadth of SAM's learned representations and the potential benefits of minimal task-specific adaptation.

3 Literature Review

3.1 Foundation Models for Computer Vision

The development of foundation models has transformed natural language processing and is increasingly influencing computer vision. These large-scale models are trained on broad data and can be adapted to numerous downstream tasks. SAM builds on the vision transformer (ViT) paradigm, combining a ViT-based image encoder pretrained with masked autoencoding (He et al., 2022) with a flexible prompt encoder and lightweight mask decoder designed for real-time inference (Kirillov et al., 2023).

3.2 Promptable Segmentation

SAM introduces promptable segmentation, where users provide flexible input guidance to specify objects of interest. Such guidance can include points, bounding boxes, or masks. This interactive paradigm enables zero-shot generalization to novel objects and scenarios without retraining (Kirillov et al., 2023). Research shows that bounding box prompts generally provide more consistent results than point prompts for complex scenes (Chen et al., 2023).

3.3 Domain Adaptation and Cityscapes Benchmark

While foundation models demonstrate impressive zero-shot capabilities, domain-specific adaptation often improves performance on specialized tasks. Training lightweight prediction heads on frozen features offers computational efficiency while preserving pretrained representations (Ranftl et al., 2021). The Cityscapes dataset has become a standard benchmark for evaluating segmentation on urban driving scenarios, with fine-grained annotations across 30 semantic classes spanning 50 cities with varying conditions (Cordts et al., 2016). Recent state-of-the-art approaches achieve mean Intersection over Union (mIoU) scores exceeding 80% (Yuan et al., 2021; Chen et al., 2022), while SAM's zero-shot performance on specialized domains reveals both strengths and limitations of web-scale training (Kirillov et al., 2023).

4 Methods

We conducted two experiments with the Segment Anything Model (SAM) and the Cityscapes dataset. First, we evaluated SAM's zero-shot performance, which showed how well SAM performs on a task without any task-specific background knowledge. Second, we evaluated our own Cityscapes-tuned semantic head on top of SAM, which showed how well domain adaptation can help SAM on downstream tasks. We then compared the two sets of results using standard evaluation metrics.

4.1 Segment Anything Model (SAM)

SAM consists of an image encoder, a prompt encoder, and a mask decoder. The image encoder produces a dense embedding of the input image. The prompt encoder embeds user guidance (point prompts or box prompts). The mask decoder predicts one or more candidate masks along with confidence scores. In all experiments, we used pretrained weights (SAM ViT-H checkpoint from the SAM Github: sam_vit_h_4b8939.pth) for inference. For the zero-shot SAM experiment, we used the image encoder, the prompt encoder, and the mask decoder, but, for the tuned semantic head experiment, we only used the image encoder as a frozen feature extractor.

4.2 Dataset

We used the Cityscapes urban driving dataset for all experiments. The Cityscapes dataset includes high-quality pixel-level annotations of 5,000 frames and an additional 20,000 weakly-annotated frames of urban driving scenes from 50 different cities. The dataset was hand-annotated on 19 different classes (road, sidewalk, various vehicles, people, vegetation, etc.). Although the annotations are on 19 classes, we restricted our experiments to five key classes (sidewalk, road, person, vehicle, and building) to keep it simple. We only used two sets of images and annotations: leftImg8bit_trainvaltest.zip and gtFine_trainvaltest.zip, which we downloaded from the Cityscapes website (cityscapes-dataset.com). The former includes the 5,000 RGB images, and the latter includes the actual annotations (ground truth) for the train/val splits, provided in different formats. We used the per-pixel label maps (...labelIds.png) for training and evaluation. Of these 5,000 images, 2,975 are set aside for training, 500 for validation, and 1,525 for testing (which do not come with publicly available ground truth annotations). Cityscapes provides dense semantic labels for street scenes, making it a useful benchmark for testing SAM's general-purpose segmentation on urban driving scenes.

Figure 4.1: Example of Cityscapes fine annotation of all classes (from the Cityscapes website)

4.3 Preprocessing

All images were loaded in RGB and passed through a preprocessing step to convert to the input that the official SAM predictor expects: 1024x1024 resolution and normalized with SAM's mean and standard deviation (restored back to Cityscapes' resolution for evaluation). For the zero-shot experiment, in order to simulate user input, we had to engineer prompts (points or bounding boxes) to feed into SAM. We generated these prompts directly from the Cityscapes ground truth annotations so that we could attempt to isolate SAM behavior from detection errors.

4.4 Issues with Point Prompts

We initially intended to use point prompts to evaluate SAM's zero-shot performance, as this method closely simulates an interactive or gaze based segmentation scenario. However, we found that generating consistent results was challenging and not repeatable for dense semantic mapping. The biggest difficulty was because of single point ambiguity in complex urban scenes. A point on a large region like roads or a building could result in a variety of valid segmentation options or objects. This meant inconsistent masks. Also, generating a large number of diverse and well placed points from ground truth that correctly isolated the necessary objects was difficult to try to engineer for a dataset the size of Cityscapes. We determined that the inherent ambiguity of single point prompting as well as the engineering challenges, did not give us a reliable baseline for the semantic segmentation task. Below is an example of point prompts being too ambiguous to provide reliable results on classes with large surface areas, such as buildings.

Figure 4.2: When prompted with a point on the building, SAM only segments the window.

4.5 Zero-shot Evaluation with Box Prompts

Our final design for zero-shot SAM evaluation uses box prompts. Using the ground truth masks, we generated tight, axis-aligned bounding boxes around each connected region of a class. These box prompts proved a much more useful, consistent, and repeatable input into SAM than point prompts. We then prompted SAM with the RGB image and the box prompt and SAM returned candidate masks, from which we selected the mask with the highest confidence score. Although this setup relies heavily on quality prompting, it measures how well SAM can produce object masks when given a reasonably informative prompt, without any fine tuning on the intended specific task.

4.6 Cityscapes-Tuned Semantic Head on Top of SAM

To move past prompt-based instance masks, we trained a lightweight semantic segmentation head on top of SAM's image features. The goal was to test whether SAM's learned image representation can support task-specific semantic labeling with minimal additional training and whether this new model can improve upon the zero-shot SAM performance.

First, we ran the SAM image encoder on each input image to obtain a dense feature embedding. We only used SAM as a frozen feature extractor to keep training practical and stable for our computational resources (an RTX 3080 GPU). A small prediction head was trained to map SAM features to Cityscapes semantic classes. The simple architecture starts with two 3x3 convolutional layers with batch normalization and ReLU activation. Then, we had a 1x1 convolutional layer to the six logits and then bilinear upsample. We used the Adam Optimizer, a learning rate of 1e-3, and cross-entropy loss. Anything not recognized as one of the five key classes was treated as a sixth class: background/void (painted black when visualized but not included in evaluation).

4.7 Evaluation Metrics

We evaluated segmentation quality using standard metrics. We calculated intersection over union (IoU) for each class and then the mean intersection over union (mIoU) over all classes. We used these metrics to compare the outputs of each experiment to the ground truth segmentations.

5 Discussion/Results

We conducted two experiments to evaluate the Segment Anything Model (SAM) in the context of urban driving scene segmentation using the Cityscapes dataset. Run 0 established a zero-shot baseline using instance level box prompts, and Run 2 implemented minimal domain adaptation by training a lightweight semantic segmentation head on SAM's frozen image features

5.1 Quantitative Analysis

The primary finding is a significant performance gain realized from minimal task specific adaptation (Table 5.2, Figure 5.3). The zero-shot baseline achieved an mIoU of 0.7192. By freezing SAM's image encoder and training only a lightweight two layer convolutional head (Run 2), the performance improved substantially to an mIoU of 0.8013, representing an 8.21 percentage point increase in the overall accuracy. Analyzing the results per class (Table 5.2), the largest gains were observed in classifying extensive regions. Buildings improved from 0.6944 to 0.8535, cars improved from 0.7374 to 0.8731, and roads improved from 0.8544 to 0.9256. This highlights that SAM's features are highly effective when coupled with a semantic head trained for global scene classification.

5.2 Interpretation of Limitations

The success of the lightweight head confirms that SAM's ViT-H image encoder, which is pretrained on the massive SA-1B dataset, learned extremely robust and generalized visual features suitable for complex urban scenes. The adaptation process efficiently closed the gap between SAM's inherent instance segmentation task and the required semantic segmentation task of Cityscapes, a classification difference that the zero shot baseline couldn't overcome. This adaptation was computationally efficient, requiring only approximately four hours of training time on a single RTX 3080 GPU for 5 epochs. A key limitation was the difficulty in using point prompts for the zero shot baseline. The ambiguity of a single point in dense urban scenes did not allow for a consistent mapping to semantic classes, which forced us to rely on ideal box prompts derived from ground truth.

Future work should focus on investigating more sophisticated semantic heads, such as employing multiscale feature fusion to further improve challenging boundaries like sidewalks and people. Also, exploring lightweight fine tuning (LoRA) of SAM's image encoder to see if minor adjustments to the general features can unlock performance gains without retraining the entire model.

Run	Batch size	Epochs	Max train samples	Max val samples	Time	mIoU
RUN 1	1	3	800	200	~1 hour	0.7495
RUN 2	2	5	2975	500	~4 hours	0.8013

Table 5.1: Semantic Head Training Experiments

Class	**RUN 0: box-prompt zero-shot (IoU)**	**RUN 2: semantic head (IoU)**
road	0.8544	0.9256
sidewalk	0.6984	0.6990
building	0.6944	0.8535
person	0.6114	0.6555
car	0.7374	0.8731
mIoU	**0.7192**	**0.8013**

Table 5.2: Zero-shot vs. Cityscapes-tuned head results table

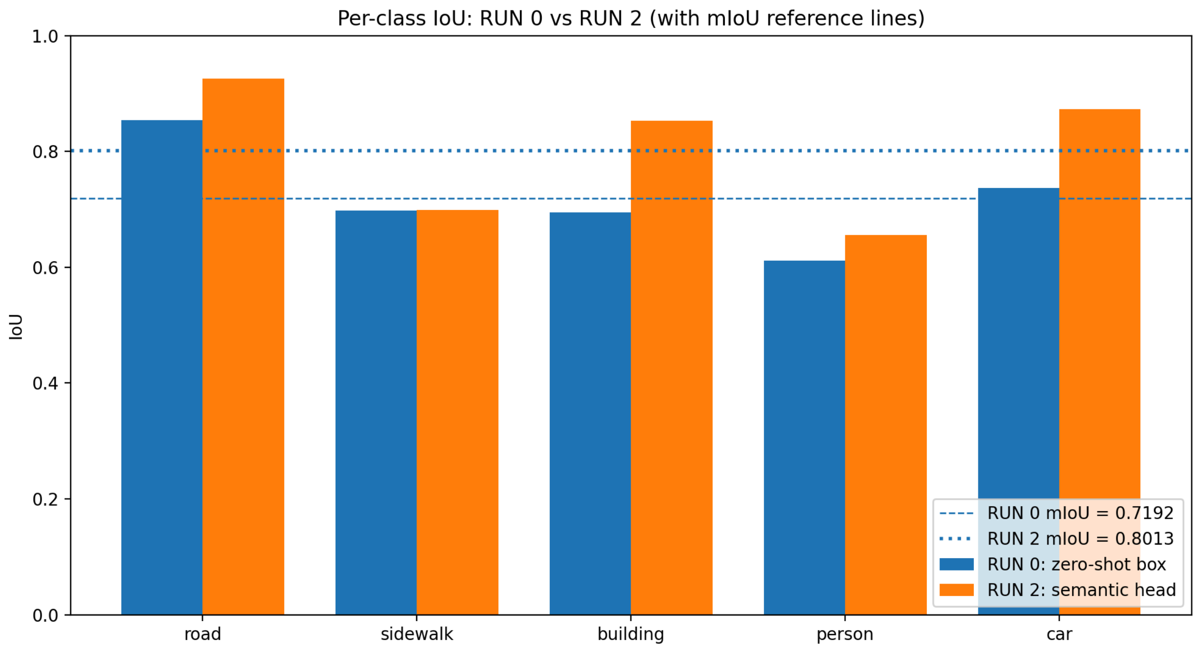

Figure 5.1: Zero-shot vs. Cityscapes-tuned head results bar graph

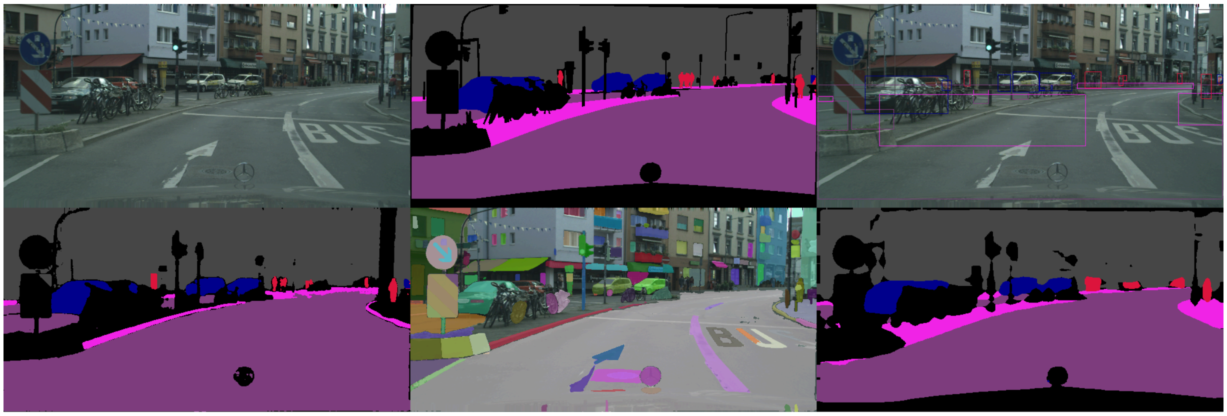

Figure 5.2: Qualitative Results (top left: original Cityscapes image, top middle: ground truth annotations, top right: generated box prompts, bottom left: zero-shot SAM output, bottom middle: SAM image encoder output, bottom right: Cityscapes-tuned semantic head output).

6 Conclusion

In conclusion, we experimented with segmentation on the Cityscapes dataset using the Segment Anything Model (SAM) with two different approaches. First, we tested SAM's zero-shot capabilities using box prompts. Second, we used SAM as a frozen feature extractor and built a small semantic Cityscapes-tuned head trained on the Cityscapes dataset. Our first approach achieved 0.7192 mean intersection over union, while our second approach achieved 0.8013 mean intersection over union, a vast improvement. This improvement does seem to prove the value of domain adaptation for the base Segment Anything Model, but it also suggests that the limiting factor for SAM is quality prompting. Our generated box prompts improved performance greatly over our point prompts, but they still caused issues such as overlapping cars only being granted a single bounding box. This suggests two paths when using SAM for specific tasks: either perfect the prompt generation or take the path we took and train on top of SAM's image encoder.

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